

Testing the Portfolio Analysis Pack at a desk

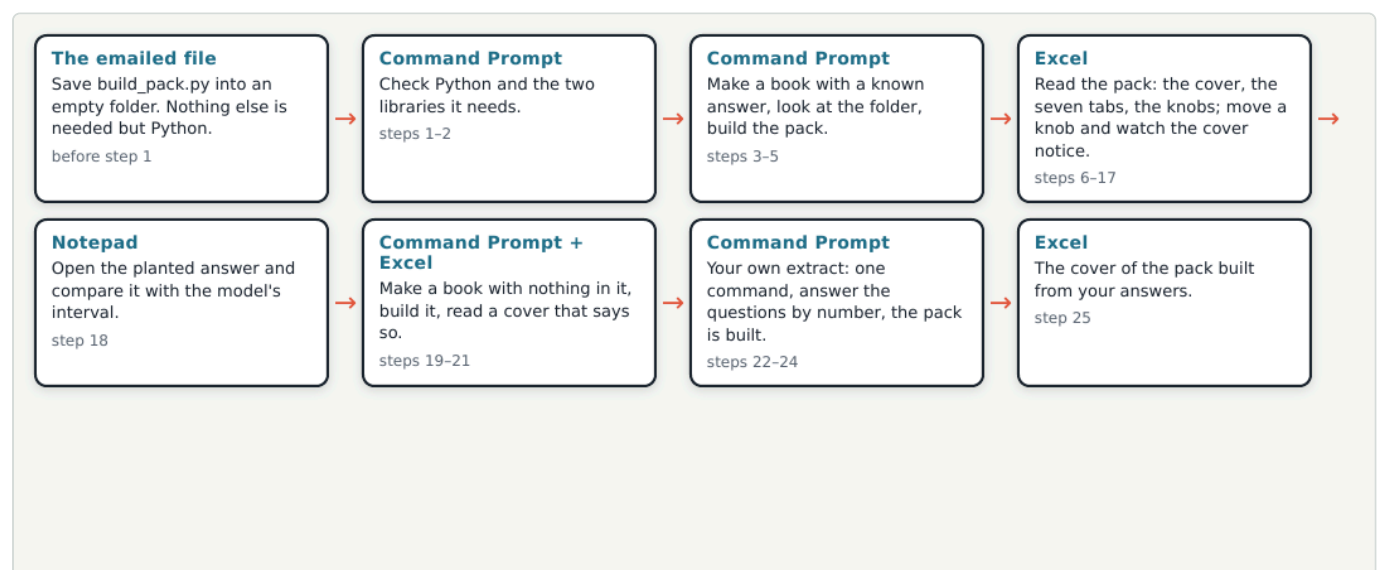
From the emailed file to a workbook you have read, in twenty-five steps. Written for a person doing it for the first time, with a picture of every screen and what a correct one looks like.

This is the procedure for testing the pack on a machine that has never seen it: a file arrives by email, you run it on a book of made-up loans with a known answer, you read the workbook it builds, and then you point it at an extract of your own. It was walked three times on 22 September 2026: on the script as first written, which found eleven things the screens said wrong (`WALKTHROUGH-DEFECTS.md`); on the patched script, after the firm asked for everything wrong to be fixed before anything shipped; and once more after Part E became a picker, so that nothing is edited by hand at the desk. Every screen below is from the last walk. Where a screen of yours differs from the picture, either the product has changed or something went wrong, and finding out which is the point of doing it again.

Two things to know before you start. The terminal pictures show a prompt that reads `desk>` ; on your machine it will show the folder you are in. And the spreadsheet pictures were taken with LibreOffice, because the machine the walk ran on has no Excel; the numbers and words are the same, the fonts are not.

You need: the emailed file `build_pack.py` , Python 3.10 or later, the two libraries it uses (`openpyxl` and `PyYAML` ; the script says which pip line to run if either is missing), and Excel. Nothing is edited by hand.

The route



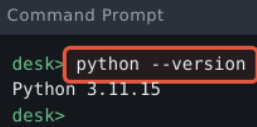
The route: eight screens in order, with what happens on each.

Make a new, empty folder, save `build_pack.py` into it, and open a Command Prompt in that folder (in File Explorer, type `cmd` in the address bar and press Enter). Every command below is typed there.

Part A · Is the machine ready?

Step 1 — Check Python

Do: type `python --version` and press Enter.



```
Command Prompt
desk> python --version
Python 3.11.15
desk>
```

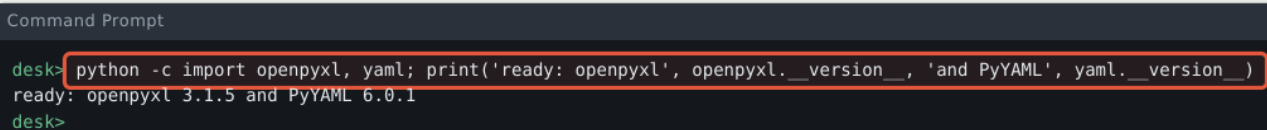
Step 1

Right when: it answers with a version of 3.10 or later. Here it was 3.11.15. If it says Python is not recognised, install it from python.org and tick "Add to PATH".

Step 2 — Check the two libraries

Do: type this exactly, on one line, and press Enter:

```
python -c "import openpyxl, yaml; print('ready: openpyxl', openpyxl.__version__, 'and PyYAML', yaml.__version__)"
```



```
Command Prompt
desk> python -c import openpyxl, yaml; print('ready: openpyxl', openpyxl.__version__, 'and PyYAML', yaml.__version__)
ready: openpyxl 3.1.5 and PyYAML 6.0.1
desk>
```

Step 2

Right when: it prints `ready:` with two version numbers. If it prints an error naming a module, run `pip install openpyxl PyYAML` and try again.

Part B · A book with a known answer

The script can make itself a book of 40,000 made-up loans in which a real effect has been planted, and write down what it planted. That is what you test on first: a book where you know the answer.

Step 3 — Make the book

Do: type `python build_pack.py --synth demo` and press Enter.

```
Command Prompt
desk> python build_pack.py --synth demo
wrote demo/loans.csv, config.yaml, planted.json (effect, seed 20260918, 40,000 loans)
then: python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
desk>
```

Step 3

Right when: the first line reads *wrote demo/loans.csv, config.yaml, planted.json (effect, seed 20260918, 40,000 loans)* and the second, *then:*, gives the build command, which is step 5's. Nothing else is printed. The planted answer is in `demo\planted.json`, and step 18 comes back to it.

Step 4 — Look at what appeared

Do: type `dir` (the picture shows `ls`, the same thing on the machine the walk ran on) and look at the folder.

```
Command Prompt
desk> ls -l . demo
.:
build_pack.py
demo

demo:
config.yaml
loans.csv
planted.json
desk>
```

Step 4

Right when: the folder holds `build_pack.py` and a `demo` folder with three files in it, and nothing else. The script unpacks itself into a temporary folder while it runs and removes it when it finishes.

Step 5 — Build the pack

Do: type this on one line and press Enter. It takes a few seconds.

```
python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
```

Command Prompt

```
desk> python build_pack.py --data demo/loans.csv --asof 2026-06-30 --config demo/config.yaml -o demo/pack.xlsx
wrote demo/pack.xlsx (165,150 bytes, sha256 908d6821cb7a46ee...)
read 0.2s · population 1.6s · models 1.5s · workbook 0.7s
29,343 seasoned loans, 257 events (event), 10,657 unseasoned excluded; gradient reads: monotonic increasing
step 4 size_band (edges): survives
step 4 amount_band (edges): survives
step 4 category_2 (levels): survives
step 6 M1: flag odds ratio 3.73 [2.84, 4.89] on 211 events over 11 coefficients, about 19 events per coefficient
step 6 M2: flag odds ratio 3.74 [2.85, 4.91] on 211 events over 21 coefficients, about 10 events per coefficient
formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1389 checks written to
_check
run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open demo/pack.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 5

Right when: the first line reads *wrote demo/pack.xlsx*, and the summary under it has these numbers: 29,343 seasoned loans, 257 events, *gradient reads: monotonic increasing*, three step-4 lines each ending *survives*, and M1's odds ratio 3.73 with the interval [2.84, 4.89] on 211 events, about 19 events per coefficient. The last three lines say that the formula check runs when Excel opens the file, that no run date was given so the pack will say so rather than guess one (add `--run-date 2026-09-22`, with the day you ran it, if you want it stamped), and what to do next. Keep this screen in mind; the cover in the next step must say the same things.

Part C · Read the workbook

Open `demo\pack.xlsx` in Excel. If Excel asks whether to enable editing or to recalculate, say yes. Read the tabs left to right; each of the next twelve steps is one tab or one thing on it.

Step 6 — The cover

Do: read the tab called *Cover*, top to bottom.

Portfolio Analysis Pack

synthetic_effect · run date not given · as-of 2026-06-30 · generator 0.1.0

The question

Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that real or something else in disguise?

The answer, in three lines

Gradient: yes. As the ratio rises, the event rate rises at every step.

Survives or collapses (step 4), for event: `size_band` (edges) — survives; `amount_band` (edges) — survives; `category_2` (levels) — survives; Model (step 6): the flag's odds ratio is 3.73 [2.84, 4.89] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added (M2) 211 events.

Regardless of the above

In 22.2% of the 25,106 seasoned loans where both `field_a` and `field_b` are present, `field_a + field_b > 1`.

`field_a`: decision one

`field_b`: decision two

Nothing in the process reacts to it.

What it rests on

29,343 seasoned loans, 257 events (event). 10,657 loans younger than 24 months are excluded from every rate.

1389 of 1389 formula checks agree (see `_check`)

1389 of 1389 formula checks agree (see `_check`)

the last line of the cover

Step 6

Right when: the black banner reads *run date not given*, because step 5 did not pass one. *The question* reads *Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that real or something else in disguise?* Under *The answer, in three lines* the three lines say what step 5's screen said: *Gradient: yes, survives three times (for event: `size_band` (edges) — survives; ...), and 3.73 [2.84, 4.89].* The last line reads *1389 of 1389 formula checks*

agree (see `_check`): every number Python wrote has been recomputed by the workbook's own formulas and matched. If that line shows a smaller first number, or the sentence about knobs having been moved (step 17 shows it), stop and say so.

Step 7 — Step 1, capture

Do: read the tab `1_Capture`.

Step 1 — Capture

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 1, capture. Before a field is trended it has to exist. For every origination quarter: loans originated, how many are still younger than the 24-month window at the as-of date, how many carry a blank in field_a, how many in field_b, and how many carry both. Blank is a state, not dirt: a quarter where a field was rarely captured is a finding about the field, and any trend in that quarter is a trend in capture, not in behaviour. Values the pack will not accept never reach this tab; the build refuses on them and lists the loans.

Quarter	Loans	Unseasoned	field_a blank	share	field_b blank	share	Both present	share
2019Q1	1,370	0	64	4.7%	610	44.5%	728	53.1%
2019Q2	1,370	0	78	5.7%	104	7.6%	1,194	87.2%
2019Q3	1,291	0	76	5.9%	99	7.7%	1,123	87.0%
2019Q4	1,386	0	74	5.3%	102	7.4%	1,210	87.3%
2020Q1	1,377	0	70	5.1%	121	8.8%	1,197	86.6%
2020Q2	1,237	0	79	6.4%	106	8.6%	1,057	85.4%
2020Q3	1,352	0	56	4.1%	132	9.8%	1,171	86.6%
2020Q4	1,306	0	57	4.4%	103	7.9%	1,147	87.8%
2021Q1	1,319	0	56	4.2%	107	8.1%	1,161	88.0%
2021Q2	1,327	0	70	5.3%	125	9.4%	1,140	85.9%
2021Q3	1,331	0	68	5.1%	105	7.9%	1,163	87.4%
2021Q4	1,337	0	70	5.2%	109	8.2%	1,168	87.4%
2022Q1	1,339	0	72	5.4%	108	8.1%	1,162	86.8%
2022Q2	1,342	0	75	5.6%	111	8.3%	1,166	86.9%
2022Q3	1,344	0	64	4.8%	98	7.3%	1,186	88.2%
2022Q4	1,342	0	66	4.9%	95	7.1%	1,185	88.3%
2023Q1	1,326	0	64	4.8%	111	8.4%	1,157	87.3%
2023Q2	1,284	0	78	6.1%	101	7.9%	1,115	86.8%
2023Q3	1,358	0	59	4.3%	112	8.2%	1,190	87.6%
2023Q4	1,311	0	59	4.5%	112	8.5%	1,149	87.6%
2024Q1	1,327	0	75	5.7%	114	8.6%	1,144	86.2%
2024Q2	1,367	0	72	5.3%	102	7.5%	1,198	87.6%
2024Q3	1,344	1,344	69	5.1%	109	8.1%	1,172	87.2%
2024Q4	1,283	1,283	77	6.0%	94	7.3%	1,117	87.1%
2025Q1	1,324	1,324	59	4.5%	94	7.1%	1,172	88.5%
2025Q2	1,370	1,370	72	5.3%	105	7.7%	1,205	88.0%
2025Q3	1,363	1,363	70	5.1%	112	8.2%	1,185	86.9%
2025Q4	1,379	1,379	58	4.2%	106	7.7%	1,220	88.5%
2026Q1	1,275	1,275	50	3.9%	94	7.4%	1,137	89.2%

Quarter	Loans	Unseasoned	field_a blank	share	field_b blank	share	Both present	share
2019Q1	1,370	0	64	4.7%	610	44.5%	728	53.1%
2019Q2	1,370	0	78	5.7%	104	7.6%	1,194	87.2%
2019Q3	1,291	0	76	5.9%	99	7.7%	1,123	87.0%
2019Q4	1,386	0	74	5.3%	102	7.4%	1,210	87.3%

the first four quarters, every column

Step 7

Right when: one row per origination quarter from 2019Q1, with the loans, how many are too young to count, and how many carry a blank in each field. The made-up book has a planted capture gap: 2019Q1 shows 610 loans (44.5%) with `field_b` blank, and every later quarter is under 10%. Quarters from 2024Q3 on show all their loans as unseasoned.

Step 8 — Step 2, prevalence

Do: read the tab 2_Prevalence.

Step 2 — Prevalence

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run data

Step 2, prevalence. Two series by origination quarter, on seasoned loans only, never merged: the capture rate is the share of seasoned loans with both field_a and field_b present; the flag rate is the share of those with both present where field_a + field_b > 1. Each carries a 95% Wilson interval. The span runs from 2019Q1 to 2024Q2: a change in the flag rate across those years is not on its own a read of underwriting drift, because the years a span covers can move either field for reasons of their own.

Quarter	Seasoned loans	Both present	Capture rate	Lower	Upper	Flagged	Flag rate	Lower	Upper
2019Q1	1,370	728	53.1%	50.5%	55.8%	155	21.3%	18.5%	24.4%
2019Q2	1,370	1,194	87.2%	85.3%	88.8%	250	20.9%	18.7%	23.3%
2019Q3	1,291	1,123	87.0%	85.0%	88.7%	262	23.3%	21.0%	25.9%
2019Q4	1,386	1,210	87.3%	85.4%	89.0%	251	20.7%	18.6%	23.1%
2020Q1	1,227	1,103	89.6%	87.7%	89.7%	258	21.5%	19.4%	24.3%
2020Q2	1,237	1,057	85.4%	83.4%	87.3%	244	23.1%	20.6%	25.7%
2020Q3	1,352	1,171	86.6%	84.7%	88.3%	265	22.6%	20.3%	25.1%
2020Q4	1,306	1,147	87.8%	85.9%	89.5%	273	23.8%	21.4%	26.4%
2021Q1	1,319	1,161	88.0%	86.2%	89.7%	237	20.4%	18.2%	22.8%
2021Q2	1,327	1,140	85.9%	83.9%	87.7%	248	21.8%	19.5%	24.2%
2021Q3	1,331	1,163	87.4%	85.5%	89.1%	263	22.6%	20.3%	25.1%
2021Q4	1,337	1,168	87.4%	85.5%	89.0%	252	21.6%	19.3%	24.0%
2022Q1	1,339	1,162	86.8%	84.9%	88.5%	293	25.2%	22.8%	27.8%
2022Q2	1,342	1,166	86.9%	85.0%	88.6%	265	22.7%	20.4%	25.2%
2022Q3	1,344	1,186	88.2%	86.4%	89.9%	263	22.2%	19.9%	24.6%
2022Q4	1,342	1,185	88.3%	86.5%	89.9%	269	22.7%	20.4%	25.2%
2023Q1	1,326	1,157	87.3%	85.4%	88.9%	251	21.7%	19.4%	24.2%
2023Q2	1,284	1,115	86.8%	84.9%	88.6%	250	22.4%	20.1%	25.0%
2023Q3	1,358	1,190	87.6%	85.8%	89.3%	246	20.7%	18.5%	23.1%
2023Q4	1,311	1,149	87.6%	85.8%	89.3%	263	22.9%	20.6%	25.4%
2024Q1	1,327	1,144	86.2%	84.2%	88.0%	240	21.0%	18.7%	23.4%
2024Q2	1,367	1,198	87.6%	85.8%	89.3%	277	23.1%	20.8%	25.6%

Quarter	Seasoned loans	Both present	Capture rate	Lower	Upper	Flagged	Flag rate	Lower	Upper
2019Q1	1,370	728	53.1%	50.5%	55.8%	155	21.3%	18.5%	24.4%
2019Q2	1,370	1,194	87.2%	85.3%	88.8%	250	20.9%	18.7%	23.3%
2019Q3	1,291	1,123	87.0%	85.0%	88.7%	262	23.3%	21.0%	25.9%
2019Q4	1,386	1,210	87.3%	85.4%	89.0%	251	20.7%	18.6%	23.1%

the first four quarters: capture rate and flag rate, each with its interval

Step 8

Right when: two rates per quarter, each with a lower and upper bound: the capture rate (both fields present) and the flag rate (the rule fires). Here the flag rate sits between 20% and 26% every quarter, and the table stops at 2024Q2, the last seasoned quarter.

Step 9 — Step 3, the gradient table

Do: read the tab *3_Gradient*, the table on the left.

Step 3 — Gradient

synthetic effect - source: loans.csv - sha256:19a1c61521c4ce9... - 29,343 seasoned loans - 257 events - generator 0.1.0 - run date not given

Step 3: gradient: Rate value = rate, a divided by field rate = 1, computed for the 29,343 seasoned loans where both fields are present (4,237 seasoned loans have a blank in one or both and are outside this table). Buckets are left-closed at the edges 0.5, 1, 2, 5; the bottom and top buckets are open. The base row is the 19,531 loans where the rule does not fire (field_a = field_b > 1). Each rate carries a 95% Wilson interval. Gap is the bucket rate minus the base rate in percentage points; multiple is the bucket rate divided by the base rate. Monotonic is real on the point estimates, comparing each pair of adjacent buckets that both hold loans. Columns headed chart: feed the chart beside the table and are not results.

Bucket	Loans	Events	Rate	Lower	Upper	Gap (pts)	Multiple	Rate change vs nearest bucket above with loans
< 0.5	12,536	39	0.31%	0.23%	0.42%	-0.22	0.59	
0.5 - 1	6,995	64	0.91%	0.72%	1.17%	0.39	1.73	0.0060
1 - 2	4,013	61	1.52%	1.19%	1.95%	0.99	2.88	0.0061
2 - 5	1,449	41	2.83%	2.09%	3.82%	2.30	5.37	0.0131
≥ 5	113	6	5.31%	2.46%	11.10%	4.78	10.07	0.0248
Base: rule does not fire	19,531	103	0.53%	0.44%	0.64%			

Monotonic?

monotonic increasing

Adjacent pairs whose intervals do not overlap

3

Interval clear of that bucket

chart bar up

chart bar down

chart: last rate

chart: last lower

chart: last upper

0.0011

0.0018

0.0034

0.0023

0.0042

0.0026

0.0043

0.0033

0.0152

0.0119

0.0195

0.0099

0.0074

0.0283

0.0209

0.0382

0.0079

0.0086

0.0373

0.0246

0.1110

event: rate by bucket, with intervals

Bucket	Loans	Events	Rate	Lower	Upper	Gap (pts)	Multiple	Rate change vs nearest bucket above with loans
< 0.5	12,536	39	0.31%	0.23%	0.42%	-0.22	0.59	
0.5 - 1	6,995	64	0.91%	0.72%	1.17%	0.39	1.73	0.0060
1 - 2	4,013	61	1.52%	1.19%	1.95%	0.99	2.88	0.0061
2 - 5	1,449	41	2.83%	2.09%	3.82%	2.30	5.37	0.0131
≥ 5	113	6	5.31%	2.46%	11.10%	4.78	10.07	0.0248
Base: rule does not fire	19,531	103	0.53%	0.44%	0.64%			

Monotonic?

monotonic increasing

Adjacent pairs whose intervals do not overlap

3

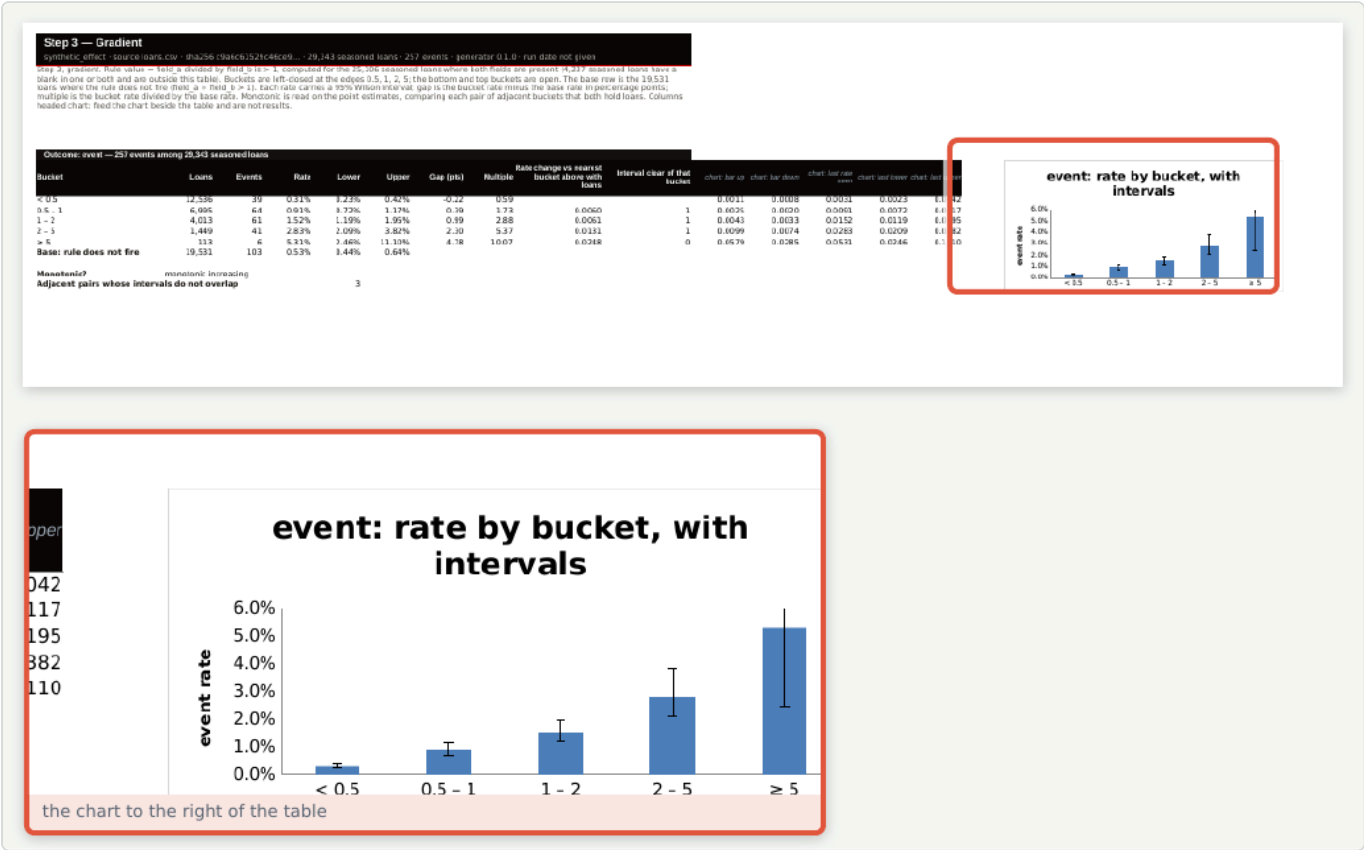
the bucket table and the two lines under it

Step 9

Right when: five buckets of the ratio from < 0.5 to ≥ 5 and a base row. The rate climbs down the column: 0.31%, 0.91%, 1.52%, 2.83%, 5.31%. Under the table, *Monotonic?* reads *monotonic increasing* and *Adjacent pairs whose intervals do not overlap* reads 3. The two long headings wrap inside their columns. The columns headed *chart*: further right feed the chart and are not results; the tab's note says so.

Step 10 — Step 3, the chart

Do: on the same tab, look at the chart to the right.



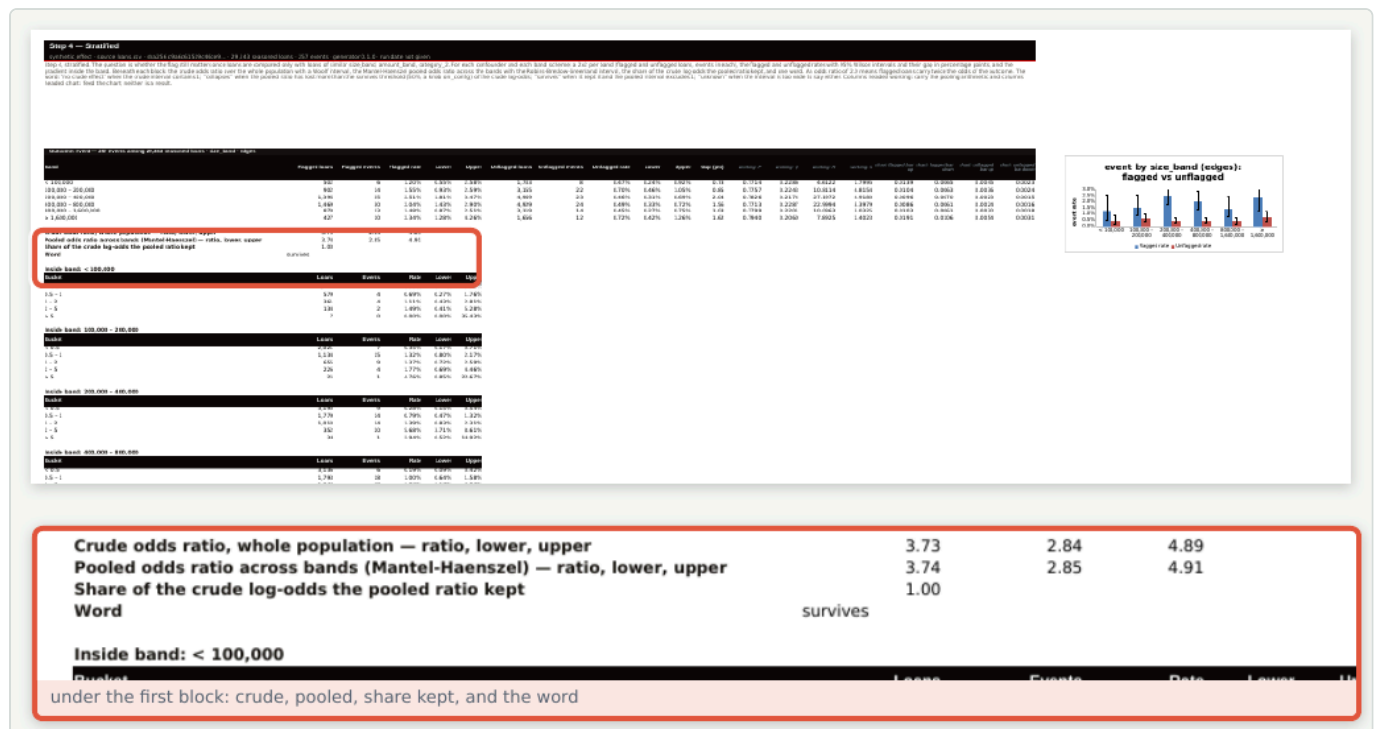
Step 10

Right when: five bars rising left to right, each with a whisker for its interval; the last bar's whisker is the widest, because that bucket holds only 113 loans.

Step 11 — Step 4, stratified

Do: read the tab 4_Stratified, the four short rows under the first block (they repeat

under every block).



Step 11

Right when: *Crude odds ratio, whole population — ratio, lower, upper 3.73, 2.84, 4.89; Pooled odds ratio across bands (Mantel-Haenszel) — ratio, lower, upper 3.74, 2.85, 4.91; Share of the crude log-odds the pooled ratio kept 1.00; and Word reads survives.* The labels read whole. The same word appears under all three blocks and matches the cover.

Step 12 — Step 5, decomposition

Do: read the tab 5_Decomposition, the first block.

Step 5 — Decomposition

synthetic_effect · source_loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run_date not given

Step 5, decomposition. Where the flag's effect sits: for each dimension listed in the question file, the flagged and unflagged rates per level with 95% Wilson intervals, the gap in percentage points, the multiple, and the share of all flagged events that fall in the level. Rows are sorted by flagged events, most first, so a pattern that is all in one level is read off the top row. A dimension with a band scheme uses the scheme's first set of edges.

Outcome: event — 257 events among 29,343 seasoned loans · by category_1

category_1	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
south	1,118	28	2.50%	1.74%	3.60%	3,992	28	0.70%	0.49%	1.01%	1.80	25.9%
east	1,174	25	2.13%	1.45%	3.12%	3,960	16	0.40%	0.25%	0.66%	1.73	23.1%
north	1,095	20	1.83%	1.19%	2.80%	3,879	20	0.52%	0.33%	0.80%	1.31	18.5%
west	1,056	20	1.89%	1.23%	2.91%	3,870	17	0.44%	0.27%	0.70%	1.45	18.5%
central	1,132	15	1.33%	0.80%	2.17%	3,830	22	0.57%	0.38%	0.87%	0.75	13.9%

Outcome: event — 257 events among 29,343 seasoned loans · by amount_band

amount_band	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
60,000 - 250,000	3,483	66	1.95%	1.54%	2.47%	12,218	68	0.56%	0.44%	0.70%	1.40	63.0%
< 60,000	1,084	21	1.94%	1.27%	2.94%	3,737	15	0.40%	0.24%	0.66%	1.54	19.4%
≥ 250,000	1,008	19	1.88%	1.21%	2.93%	3,576	20	0.56%	0.36%	0.86%	1.33	17.6%

Outcome: event — 257 events among 29,343 seasoned loans · by origination_year

origination_year	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
2023	1,010	24	2.38%	1.60%	3.51%	3,601	13	0.36%	0.21%	0.62%	2.02	22.2%
2021	1,000	23	2.30%	1.54%	3.43%	3,632	16	0.44%	0.27%	0.71%	1.86	21.3%
2020	1,040	18	1.73%	1.10%	2.72%	3,527	31	0.88%	0.62%	1.24%	0.85	16.7%
2022	1,090	17	1.56%	0.93%	2.48%	3,609	18	0.50%	0.32%	0.79%	1.06	15.7%
2019	918	14	1.53%	0.91%	2.56%	3,337	14	0.42%	0.25%	0.70%	1.11	13.0%
2024	517	12	2.32%	1.33%	4.01%	1,825	11	0.60%	0.34%	1.08%	1.72	11.1%

Outcome: event — 257 events among 29,343 seasoned loans · by category_1

category_1	Flagged loans	Flagged events	Flagged rate	Lower	Upper	Unflagged loans	Unflagged events	Unflagged rate	Lower	Upper	Gap (pts)	Share of flagged events
south	1,118	28	2.50%	1.74%	3.60%	3,992	28	0.70%	0.49%	1.01%	1.80	25.9%
east	1,174	25	2.13%	1.45%	3.12%	3,960	16	0.40%	0.25%	0.66%	1.73	23.1%
north	1,095	20	1.83%	1.19%	2.80%	3,879	20	0.52%	0.33%	0.80%	1.31	18.5%
west	1,056	20	1.89%	1.23%	2.91%	3,870	17	0.44%	0.27%	0.70%	1.45	18.5%
central	1,132	15	1.33%	0.80%	2.17%	3,830	22	0.57%	0.38%	0.87%	0.75	13.9%

the first dimension, rows sorted by flagged events

Step 12

Right when: one block per dimension listed in the question file, rows sorted by flagged events with the biggest first. For `category_1` the flagged rate sits between 1.3% and 2.5% in every row and the unflagged rate under 0.8%, so the effect is not hiding in one region.

Step 13 — Step 6, the model

Do: read the tab *6_Model*, the first row of the M1 table.

Step 6 — Model

synthetic_effect · source: loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 6, model. Two logistic regressions side by side, values only (a regression cannot be written as a cell formula, so this tab does not recalculate and its numbers have no check-tab twin). M1 has the flag plus the controls; M2 adds every confounder that is not already a control. A term's odds ratio says how the odds of the outcome move with that term, everything else held fixed: 2.0 on the flag means flagged loans carry twice the odds. The interval is the 95% range it plausibly sits in. If the flag's ratio falls from M1 to M2, what was lost is explained by the confounders. Categorical terms are measured against a reference level, the one with the most loans. A continuous control marked log enters as its natural logarithm. The tree beneath is a search for the two- or three-way combination with the most contrast, printed as rules with the loans, events and rate in each leaf; a lead to check, not a proof.

Outcome: event

Loans in the design: 25,106 seasoned loans with both rule fields and every predictor present; 0 left out for a blank predictor.

M1: flag + controls
25,106 loans · 211 events · 11 estimated coefficients · events per parameter 19.2

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.729	2.842	4.894	1.3162	0.1387	
origination_year=2021	1.164	0.736	1.843	0.1523	0.2343 against origination_year=2022	
origination_year=2023	1.104	0.694	1.758	0.0991	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M2: M1 + confounders not already controls
25,106 loans · 211 events · 21 estimated coefficients · events per parameter 10.0
Skipped: amount_band (its field amount is already a control)

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.742	2.851	4.911	1.3197	0.1387	
origination_year=2021	1.164	0.735	1.843	0.1519	0.2344 against origination_year=2022	
origination_year=2023	1.110	0.693	1.767	0.1042	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M1: flag + controls
25,106 loans · 211 events · 11 estimated coefficients · events per parameter 19.2

Term	Odds ratio	Lower	Upper	Coefficient	Std error	Reference level
flag	3.729	2.842	4.894	1.3162	0.1387	
origination_year=2021	1.164	0.736	1.843	0.1523	0.2343 against origination_year=2022	
origination_year=2023	1.104	0.694	1.758	0.0991	0.2372 against origination_year=2022	
origination_year=2020	1.453	0.939	2.249	0.3735	0.2228 against origination_year=2022	
origination_year=2019	0.909	0.551	1.498	-0.0959	0.2549 against origination_year=2022	
origination_year=2024	1.351	0.796	2.295	0.3010	0.2703 against origination_year=2022	
amount (log)	1.004	0.848	1.189	0.0041	0.0861	
category_1=south	1.397	0.931	2.096	0.3342	0.2070 against category_1=east	
category_1=north	1.019	0.657	1.580	0.0187	0.2237 against category_1=east	
category_1=central	0.936	0.598	1.463	-0.0667	0.2281 against category_1=east	
category_1=west	0.962	0.615	1.504	-0.0391	0.2282 against category_1=east	
intercept				-5.4884		

M1: the flag row is the first term; its odds ratio and interval are the answer

Step 13

Right when: the row *flag* shows odds ratio 3.729, lower 2.842, upper 4.894, the same three numbers the cover rounds to two places. The line above the table reads *211 events · 11 estimated coefficients · events per parameter 19.2*; the M2 table below it says 3.742.

Step 14 — Step 7, the observation that stands regardless

Do: read the tab `7_Control`.

Step 7 — The observation that stands regardless

synthetic_effect · source loans.csv · sha256 c9a6c61529c46ce9... · 29,343 seasoned loans · 257 events · generator 0.1.0 · run date not given

Step 7, the observation that stands regardless of what steps 3 to 6 found. It rests on the counts alone: how often the two fields disagree in the seasoned book, and what the question file says reacts to that today. The tool asserts nothing of its own about how either field is verified or used; the two lines under the share are the question file's words.

Seasoned loans with both fields present
Of those, loans where the rule fires
Share

25,106
5,575
22.2%

In 22.2% of the 25,106 seasoned loans where both field_a and field_b are present, field_a + field_b > 1.
field_a: decision one
field_b: decision two
Nothing in the process reacts to it.

Seasoned loans with both fields present
Of those, loans where the rule fires
Share

25,106
5,575
22.2%

In 22.2% of the 25,106 seasoned loans where both field_a and field_b are present, field_a + field_b > 1.
field_a: decision one
field_b: decision two
Nothing in the process reacts to it.

the counts and the sentence built from them

Step 14

Right when: three counts (25,106 loans with both fields, 5,575 where the rule fires, share 22.2%) and the sentence built from them. This tab does not depend on anything steps 3 to 6 found.

Step 15 — The knobs

Do: read the tab `_config`, the block *Live knobs*.

Live knobs — change these and the workbook recalculates

Confidence level95.00%The interval width every rate carries. 0.95 means a 95% interval.
Interval methodWilsonWilson (default) or Clopper-Pearson. Pick from the list.
Survives threshold0.50Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.
z for the confidence level1.959964Derived from the confidence level; not a knob.

Rebuild knobs — shown for the record; change the question file and build again

Question filesynthetic_effect
Rulefield_a ÷ field_b > 1
Bucket edges0.5, 1, 2, 5
Outcomeevent, from outcome_date
Window (months)24
Filternone
As-of date2026-06-30

Live knobs recalculate the workbook when changed: the confidence level and the interval method. Rebuild knobs are shown for the record and take effect only when the pack is built again.

Live knobs — change these and the workbook recalculates

Confidence level95.00%The interval width every rate carries. 0.95 means a 95% interval.
Interval methodWilsonWilson (default) or Clopper-Pearson. Pick from the list.
Survives threshold0.50Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.
z for the confidence level1.959964Derived from the confidence level; not a knob.

the live knobs

Step 15

Right when: *Confidence level 95.00%, Interval method Wilson, Survives threshold 0.50* (its note: *Step 4's word: the share of the crude effect that must remain for 'survives'.* *0.50 means half.*), and *z 1.959964*. The block under it, *Rebuild knobs*, is for the record only; its *Outcome* line reads *event, from outcome_date*, the word and the column you gave.

Step 16 — Move a knob

Do: click the cell that reads *Wilson* and pick *Clopper-Pearson* from the list it offers. (In the walk, a script wrote the cell instead of a hand; the workbook does not know the difference.)

Live knobs — change these and the workbook recalculates

Confidence level

Interval method

Survives threshold

z for the confidence level

95.00%

Clopper-Pearson

0.50

1.959964

The interval width every rate carries. 0.95 means a 95% interval.

Wilson (default) or Clopper-Pearson. Pick from the list.

Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means half.

Derived from the confidence level; not a knob.

Rebuild knobs — shown for the record; change the question file and build again

Question file

Rule

Bucket edges

Outcome

Window (months)

Filter

As-of date

synthetic_effect

field_a < field_b > 1

0.5, 1, 2, 5

event, from outcome_date

24

none

2026-06-30

Live knobs recalculate the workbook when changed: the confidence level and the interval method. Rebuild knobs are shown for the record and take effect only when the pack is built again.

Live knobs — change these and the workbook recalculates

Confidence level

Interval method

Survives threshold

z for the confidence level

95.00%

Clopper-Pearson

0.50

1.959964

The interval width every rate carries. 0.95 means a 95% interval.

Wilson (default) or Clopper-Pearson. Pick from the list.

Step 4's word: the share of the crude effect that must remain for 'survives'. 0.50 means hal

Derived from the confidence level; not a knob.

Interval method now reads Clopper-Pearson

Step 16

Right when: the cell reads *Clopper-Pearson* and every interval in the workbook recalculates; the bounds on every tab move a little.

Step 17 — The cover notices

Do: go back to the *Cover* tab and read its last line.

Portfolio Analysis Pack
synthetic_effect = 0.05; risk_score = 0.5; survival_threshold = 0.5; generated 2.2.0

The question
Survives analysis table is checked by field_0 is > 1. Is event rate condition being there than among loans where it is not, and is that real or something else in disguise?

The answer, in three lines
Gradient: yes. As the ratio rises, the event rate rises at every step.

Survives or changes index 4), for event_size band indices) — survives; amount band indices) — survives; category_2 levels) — survives
Ratio (step 4): the flag's odds ratio is 2.72 [2.64, 4.03] with the controls (M1) and 2.74 [2.85, 4.91] with the confounders added (M2), on 233 events.

Regardless of the above
In 27.2% of the 35,554 unconverted loans where both field_0 and field_1 are present, field_0 = field_1 > 1.

field_0 decides one
field_1 decides two
Nothing in the process leads to it.

What it rests on
23,342 assessed loans, 257 events (event). 16,437 loans younger than 24 months are excluded from every rate.

The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5). The Python column on _check was computed at those settings, so the formula check applies there only; set the knobs back to compare.

tools.csv | f3a13e cfa6c91237c18c05995974d02f7449142c4677313f18b1d4645a13f7517ab29 - 40,000 rows READ | QUESTION EN
synthetic_effect = 0.05; risk_score = 0.5; survival_threshold = 0.5; generated 2.2.0

Step 17

Right when: the line that read *1389 of 1389 formula checks agree* has been replaced by *The live knobs have been moved from the settings this pack was built with (confidence 95%, method Wilson, survives threshold 0.5)...* and tells you to set the knobs back to compare. Set the knob back to Wilson and the original line returns. Close the workbook without saving.

Step 18 — The planted answer

Do: open `demo\planted.json` in Notepad.

```
demo\planted.json — Notepad
1 {
2   "seed": 20260918,
3   "loans": 40000,
4   "mode": "effect",
5   "effect_per_doubling": 2.0,
6   "base_rate": 0.012,
7   "asof": "2026-06-30",
8   "window_months": 24,
9   "true_marginal_flag_odds_ratio": 4.317157028211426,
10  "note": "true_marginal_flag_odds_ratio is the odds ratio of flagged (ratio > 1) against unflagged loans computed
from the probabilities the generator drew with, before sampling noise and before seasoning; it is what the pack should
recover within its interval."
11 }
```

Step 18

Right when: *true_marginal_flag_odds_ratio* is 4.317..., and that number lies inside the model's interval from step 13, 2.84 to 4.89. The note under it says this is what the pack should recover; it did.

Part D · A book with nothing in it

The second test is the opposite one: a book in which nothing was planted, where the pack must say so.

Step 19 — Make the empty book

Do: type `python build_pack.py --synth null --null` and press Enter.

```
Command Prompt
desk> python build_pack.py --synth null --null
wrote null/loans.csv, config.yaml, planted.json (null, seed 20260918, 40,000 loans)
then: python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
desk>
```

Step 19

Right when: the first line reads *wrote null/loans.csv, config.yaml, planted.json (null, seed 20260918, 40,000 loans)*.

Step 20 — Build it

Do: type this on one line and press Enter:

```
python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
```

```
Command Prompt
desk> python build_pack.py --data null/loans.csv --asof 2026-06-30 --config null/config.yaml -o null/pack.xlsx
wrote null/pack.xlsx (164,930 bytes, sha256 e9e3b402e20775de...)
read 0.2s · population 1.6s · models 1.5s · workbook 0.7s
29,337 seasoned loans, 371 events (event), 10,663 unseasoned excluded; gradient reads: not monotonic
step 4 size_band (edges): no crude effect
step 4 amount_band (edges): no crude effect
step 4 category_2 (levels): no crude effect
step 6 M1: flag odds ratio 0.82 [0.62, 1.08] on 324 events over 11 coefficients, about 29 events per coefficient
step 6 M2: flag odds ratio 0.82 [0.62, 1.09] on 324 events over 21 coefficients, about 15 events per coefficient
formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1389 checks written to
_check
run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open null/pack.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 20

Right when: *gradient reads: not monotonic*, three step-4 lines each ending *no crude effect*, and M1's odds ratio 0.82 with an interval [0.62, 1.08] that has 1 inside it.

Step 21 — Its cover

Do: open `null\pack.xlsx` and read the three answer lines on the cover.

Portfolio Analysis Pack

synthetic_null · run date not given · as-of 2026-06-30 · generator 0.1.0

The question

Loans where `field_a` divided by `field_b` is > 1 : is event more common among them than among loans where it is not, and is that something else in disguise?

The answer, in three lines

Gradient: not one-directional. The event rate does not rise at every step of the ratio.

Survives or collapses (step 4), for event: `size_band` (edges) — no crude effect; `amount_band` (edges) — no crude effect; `category_2` (levels) — no crude effect

Model (step 6): the flag's odds ratio is 0.82 [0.62, 1.08] with the controls (M1) and 0.82 [0.62, 1.09] with the confounders added (M2) 324 events.

Regardless of the above

In 22.3% of the 25,119 seasoned loans where both `field_a` and `field_b` are present, $\text{field_a} \div \text{field_b} > 1$.

`field_a`: decision one

`field_b`: decision two

Nothing in the process reacts to it.

What it rests on

29,337 seasoned loans, 371 events (event). 10,663 loans younger than 24 months are excluded from every rate.

1389 of 1389 formula checks agree (see `_check`)

The answer, in three lines

Gradient: not one-directional. The event rate does not rise at every step of the ratio.

Survives or collapses (step 4), for event: `size_band` (edges) — no crude effect; `amount_band` (edges) — no crude effect; `category_2` (levels) — no crude effect

Model (step 6): the flag's odds ratio is 0.82 [0.62, 1.08] with the controls (M1) and 0.82 [0.62, 1.09] with the confounders added (M2) 324 events.

the three answer lines on the book with nothing planted

Step 21

Right when: *Gradient: not one-directional, no crude effect three times, and 0.82 [0.62, 1.08].* A pack that found an effect here would be wrong. Close the workbook.

Part E · Your own extract

The made-up book came with its question file. An extract of your own does not, and the tool will not guess. One command lists your columns with a number beside each and asks you, one question at a time, which is which. You answer with the numbers. It writes the question file from your answers, checks it, and builds the pack. Nothing is typed into a file. The walk did this on the made-up extract so the answers can be checked against Part B; on a real one the columns will be yours.

Step 22 — Start the picker and answer the first seven questions

Do: type `python build_pack.py --setup demo/loans.csv` and press Enter. Read the numbered list of columns. Answer each question with a number from that list (or press Enter where the question offers a default). The answers for the made-up extract: Enter (it found the one column with a different value on every row), 2, 3, 4, Enter (a ratio), Enter (fires above 1), Enter (edges 0.5, 1, 2, 5).

```
Command Prompt
desk> python build_pack.py --setup demo/loans.csv
The columns of loans.csv (40,000 rows):
 1 loan_id      text      blank 0%   e.g. L000001, L000002, L000003
 2 origination_date date     blank 0%   e.g. 2022-06-15, 2026-06-24, 2025-09-14
 3 field_a      decimal  blank 5%   e.g. 203861.84, 306113.51, 708822.85
 4 field_b      decimal  blank 9%   e.g. 1022352.32, 389558.11, 474987.17
 5 amount       decimal  blank 0%   e.g. 711237.89, 70965.26, 93816.01
 6 category_1   text     blank 0%   e.g. east, south, central
 7 category_2   text     blank 0%   e.g. kind_6, kind_5, kind_3
 8 code_1       integer  blank 0%   e.g. 523567, 919149, 814961
 9 outcome_date date     blank 99%  e.g. 2025-02-04, 2025-04-24, 2022-05-30
10 flag_1       integer  blank 0%   e.g. 0, 1
11 measure_a    decimal  blank 0%   e.g. 261612.06, 43869.14, 10749.29
12 measure_b    decimal  blank 0%   e.g. 711237.89, 70965.26, 93816.01

1. Which column numbers the loans, one value per loan? (Enter for 1 = loan_id)
>
2. Which column holds the date the loan was made?
> 2
3. The rule compares two number columns. Which is the first (the top of a ratio)?
> 3
4. Which is the second (the bottom of the ratio)?
> 4
5. Compare them as
  1 a ratio: field_a ÷ field_b
  2 a difference: field_a - field_b
(Enter for 1)
>
6. The flag fires when the ratio is above what? (Enter for 1)
>
7. Edges for the steps of the ratio in the gradient (numbers, smallest first) (Enter for 0.5, 1, 2, 5)
>
..
```

Step 22

Right when: the list shows every column with what it reads as (text, date, decimal, integer), how often it is blank and three sample values, and the first question already proposes `loan_id`, the only column that is different on every row. A wrong answer (a text column where a number is needed, a number not on the list) is refused with a reason and the question is asked again.

Step 23 — The outcome, the as-of date and the groups

Do: keep answering: 1 (a column with the date it happened), 9 (outcome_date), the word event, Enter (24 months), the as-of date 2026-06-30, then 4, 5, 7 for the columns the effect might be hiding in. For field_b type the edges 100000, 200000, 400000, 800000, 1600000; for amount, 60000, 250000 (the picker proposes the quartiles of the data if you would rather press Enter). Enter for *what reacts today* (none), Enter for *known only after* (nothing), Enter for today's date (unstamped).

Command Prompt

```
...
8. How does the extract say a loan went bad?
  1 a column with the date it happened
  2 a yes/no column the bank already set (1 = yes)
  3 a measure at the as-of date, from two number columns (e.g. utilisation)
(Enter for 1)
> 1
9. Which column has the date it happened?
> 9
The word every tab will use for the event (e.g. default, charge-off)
> event
Months a loan needs on book before it counts (Enter for 24)
>
The as-of date the pack is built at (YYYY-MM-DD)
> 2026-06-30
Step 4 tests whether the flag still matters inside groups of similar loans. Without any
group, the pack cannot tell an effect from something else in disguise.
Which columns might the effect be hiding in (size, product, region ...)? (numbers separated by commas; Enter for none) –
number columns are banded, text columns taken level by level
> 4, 5, 7
Band edges for field_b (numbers, smallest first) (Enter for 200,000, 400,000, 780,000)
> 100000, 200000, 400000, 800000, 1600000
Band edges for amount (numbers, smallest first) (Enter for 70,000, 120,000, 210,000)
> 60000, 250000
What at the bank reacts to this contradiction today? (Enter for none)
>
Of the columns in use, which were known only AFTER the loan was made (other than the outcome)? (numbers separated by
commas; Enter for none)
>
Today's date, to stamp on the pack (YYYY-MM-DD; Enter for none)
>
...
```

Step 23

Right when: the picker explains why it asks for groups (*without any group, the pack cannot tell an effect from something else in disguise*), proposes band edges for each number column from the data, and asks nothing you cannot answer from the list. If you name a column as known only after the loan was made and it is one of the rule's columns, the picker refuses it as a leak and asks again.

Step 24 — It writes the file, checks it and builds

Do: nothing; read the screen.

```
Command Prompt

...
wrote demo/question.yaml from your answers
building demo/field_a_vs_field_b.xlsx ...
wrote demo/field_a_vs_field_b.xlsx (172,064 bytes, sha256 25afd0badd7fe00f...)
  read 0.5s · population 1.8s · models 1.2s · workbook 0.8s
  29,343 seasoned loans, 257 events (event), 10,657 unseasoned excluded; gradient reads: monotonic increasing
    step 4 field_b (edges): survives
    step 4 amount (edges): survives
    step 4 category_2 (levels): survives
    step 6 M1: flag odds ratio 3.72 [2.84, 4.88] on 211 events over 6 coefficients, about 35 events per coefficient
    step 6 M2: flag odds ratio 3.74 [2.85, 4.91] on 211 events over 18 coefficients, about 12 events per coefficient
  formula check: runs when Excel opens the file (this machine has no spreadsheet engine) – 1473 checks written to
  _check
  run date: not given, so the pack says so; pass --run-date YYYY-MM-DD to stamp it
then: open demo/field_a_vs_field_b.xlsx in Excel; the cover's last line should say every formula check agrees
desk>
```

Step 24

Right when: *wrote demo/question.yaml from your answers*, then *building demo/field_a_vs_field_b.xlsx ...* (the pack is named after the two columns), then the same summary as step 5: 29,343 seasoned loans, 257 events, *monotonic increasing*, three step-4 lines each ending *survives*, M1 3.72 [2.84, 4.88], and the *then:* line telling you to open the file in Excel. M1 has 6 coefficients where step 5 had 11: the picker adds only the origination year as a control, where the made-up book's own file also holds the amount and a category.

Step 25 — Its cover

Do: open `demo\field_a_vs_field_b.xlsx` and read the three answer lines.

Portfolio Analysis Pack
field_a_vs_field_b · run date not given · as-of 2026-06-30 · generator 0.1.0

The question
Loans where field_a divided by field_b is > 1: is event more common among them than among loans where it is not, and is that something else in disguise?

The answer, in three lines
Gradient: yes. As the ratio rises, the event rate rises at every step.
Survives or collapses (step 4), for event: field_b (edges) — survives; amount (edges) — survives; category_2 (levels) — survives
Model (step 6): the flag's odds ratio is 3.72 [2.84, 4.88] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added (M2) 211 events.

Regardless of the above
In 22.2% of the 25,106 seasoned loans where both field_a and field_b are present, field_a ÷ field_b > 1.

Nothing in the process reacts to it.

What it rests on
29,343 seasoned loans, 257 events (event). 10,657 loans younger than 24 months are excluded from every rate.

1473 of 1473 formula checks agree (see _check)

Source
loans.csv · sha256 c9a6c61529c46ce99b974ad0f8c74493a2c4e7f39c9948fbd4b45a19f5d9ae29 · 40,000 rows read · question f
field_a_vs_field_b · sha256 47e8e1a30d4715ca...

The answer, in three lines
Gradient: yes. As the ratio rises, the event rate rises at every step.
Survives or collapses (step 4), for event: field_b (edges) — survives; amount (edges) — survives; category_2 (levels) — survives
Model (step 6): the flag's odds ratio is 3.72 [2.84, 4.88] with the controls (M1) and 3.74 [2.85, 4.91] with the confounders added (M2) 211 events.

the three answer lines of the pack built from your answers

Step 25

Right when: *Gradient: yes, survives three times (for event: field_b (edges) — survives; amount (edges) — survives; category_2 (levels) — survives), and 3.72 [2.84, 4.88], with 1473 of 1473 formula checks agree lower down. The banner reads field_a_vs_field_b, the name the picker gave the question from your two columns, and run date not given. The same answer as step 6, from your own answers.*

Run the command again later and it finds the question file and offers to reuse it, asking only for the as-of date.

When you are done

You have run the tool on a book with a known answer and watched it find the answer, on a book with no answer and watched it say so, and on an extract whose columns you designated yourself by number. To do the same on a real extract, run `python build_pack.py --setup YOUR-FILE.csv` (an `.xlsx` works too: add `--sheet` and the tab's name) and answer the questions for that book: its own columns, its real as-of date, today's date to stamp, and the groups that matter for it. There is nothing to tidy afterwards: the script leaves nothing beside itself but the question file and the pack.